

Computational modeling of neural populations

Part II: Neural fields — Spatially extended Wilson-Cowan model

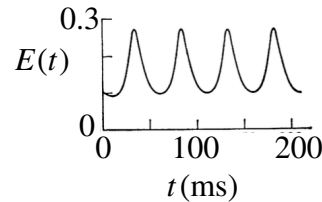
Why spatial extension?

- Localized Wilson-Cowan focuses on only temporal dynamics.

- Resting state
- Limit cycles

⋮

Limit cycles:



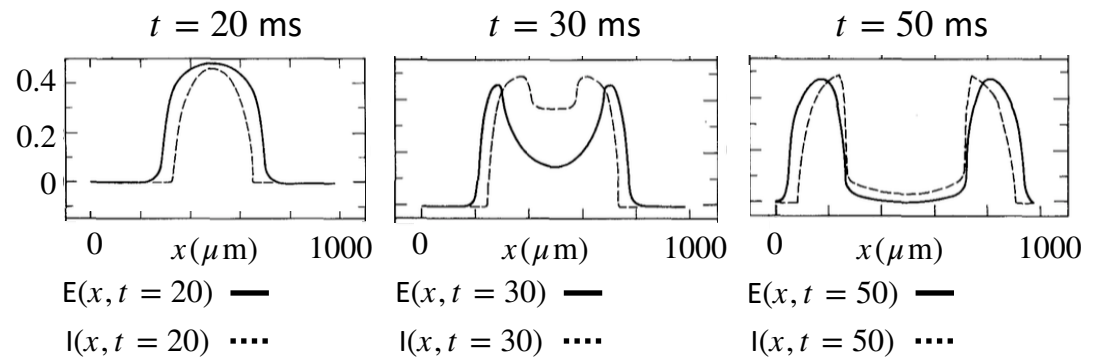
Wilson-Cowan 1972
Biophysical J

- Spatially extended Wilson-Cowan focuses on spatiotemporal dynamics.

- **Spatially** inhomogeneous steady states
- **Traveling** waves

⋮

Traveling waves:



Wilson-Cowan 1973
Kybernetik

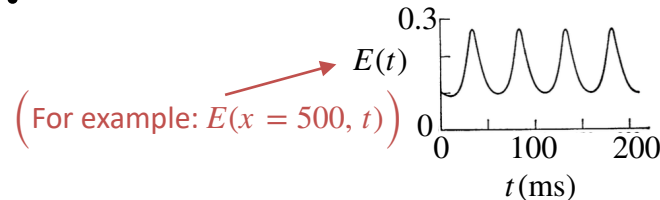
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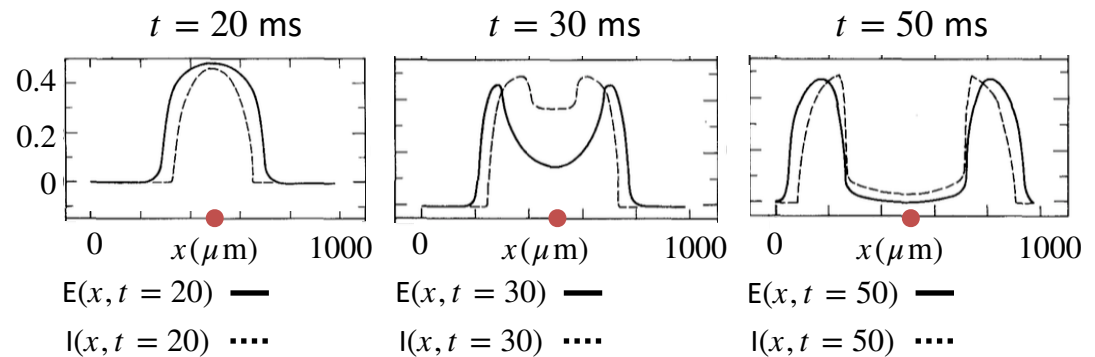
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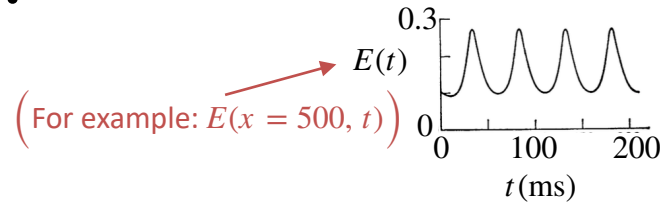
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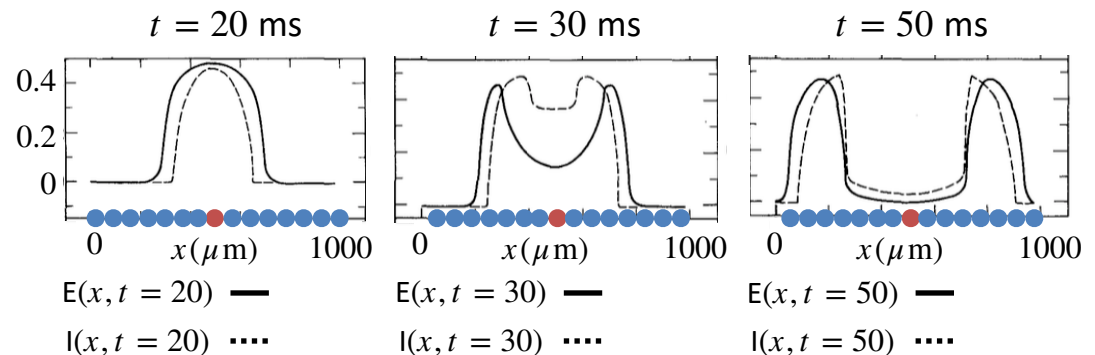
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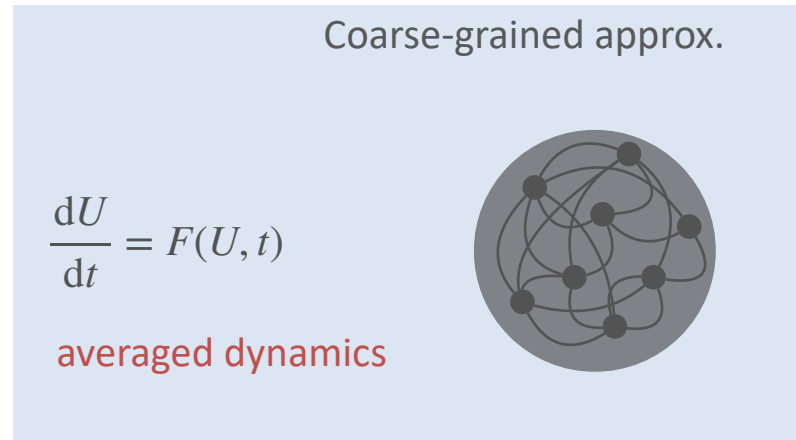
⋮

Traveling waves:



Wilson-Cowan 1973
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Spatial extension allows us to consider the neurons which are not necessarily in close proximity.



Localized neural field: $\frac{dU}{dt} = F(U, t), \quad U(t)$

- Neurons in close proximity
- No spatial dependence
- No concept of connectivity

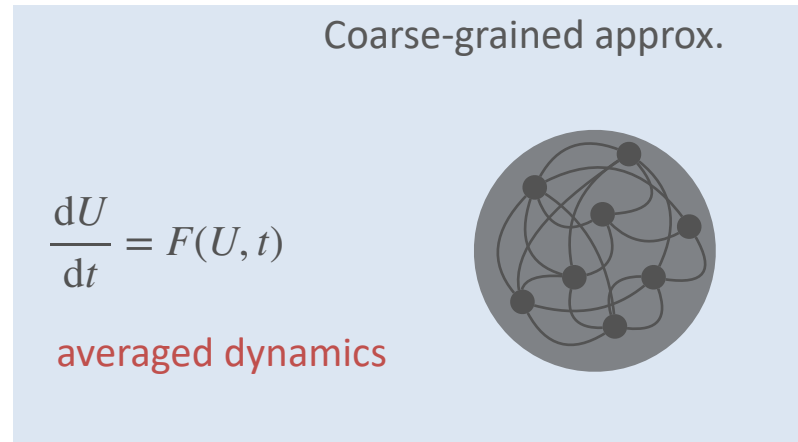
- Wilson-Cowan '72

Spatially extended neural field: $\frac{\partial U}{\partial t} = F(U, x, t), \quad U(x, t)$

- Spatially distributed neurons
- Spatial dependence
- Distance-based connectivity

- Wilson-Cowan '73

- Amari '77



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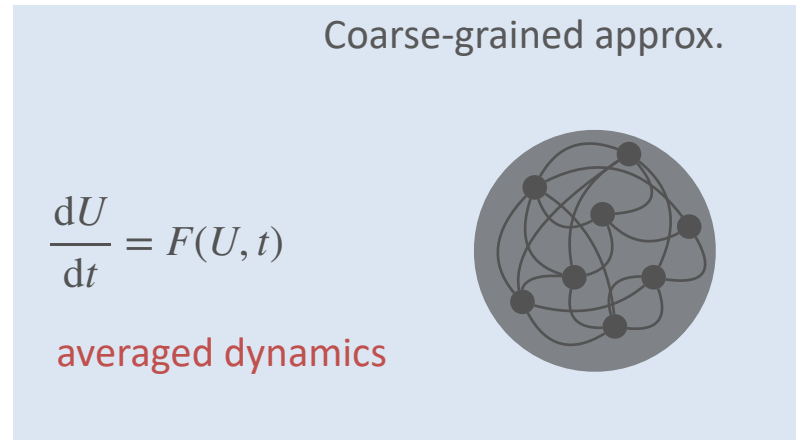
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Spatially extended neural field: $\frac{\partial U}{\partial t} = F(U, x, t), \quad U(x, t)$

- Spatially distributed neurons
- Spatial dependence
- **Distance-based connectivity**

- **Wilson-Cowan '73**

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Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

Modeling assumptions

1. There are two types of neurons, excitatory and inhibitory, and they are continuously and uniformly distributed in the cortical space.
 - Excitatory and inhibitory neurons co-exist in the cortical space, similarly to continuously distributed neurons on a sheet.
2. Neural activity can be represented in terms of proportion of activated cells.
3. Neural processes are dependent on the interaction of excitatory and inhibitory neurons.
 - Individual neurons are assumed to integrate incoming excitation-inhibition **both** spatially and temporally.
 - Influence of excitatory and inhibitory neurons on each other is mediated by **distance-dependent** kernels.
4. All possible interactions are allowed: excitatory $\rightarrow$ excitatory (ee), inhibitory $\rightarrow$ inhibitory (ii), excitatory $\rightarrow$ inhibitory (ie), inhibitory $\rightarrow$ excitatory (ei).
 - Each interaction type will be assumed to be function only of the distance (distance-dependent kernel).
 - Thus, the cortical tissue is assumed to be isotropic and homogeneous.

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

$$\mu \frac{\partial \bar{E}(x, t)}{\partial t} = -\bar{E}(x, t) + (1 - \bar{E}(x, t)) S_e \left((\bar{E} \otimes \omega_{ee})(x, t) - (\bar{I} \otimes \omega_{ei})(x, t) + \bar{P}(x, t) \right)$$

$$\mu \frac{\partial \bar{I}(x, t)}{\partial t} = -\bar{I}(x, t) + (1 - \bar{I}(x, t)) S_i \left((\bar{E} \otimes \omega_{ie})(x, t) - (\bar{I} \otimes \omega_{ii})(x, t) + \bar{Q}(x, t) \right)$$

$[-x_0, x_0] \subset \mathbb{R}$: spatial interval cortical layer ($x \in [-x_0, x_0]$)

T : positive constant final time

$\bar{E}, \bar{I} : [-x_0, x_0] \times [0, T] \rightarrow [0, 1]$ proportion of firing excitatory ($\bar{E}$) or inhibitory ($\bar{I}$)
neurons located at x and at instant t

$\bar{P}, \bar{Q} : [-x_0, x_0] \times [0, T] \rightarrow \mathbb{R}$ external stimuli

S_e, S_i : nonlinear functions population transfer functions

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Spatially extended

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Localized

$$\mu \frac{d\bar{E}(t)}{dt} = -\bar{E}(t) + (1 - \bar{E}(t)) S_e \left(w_{ee} \bar{E}(t) - w_{ei} \bar{I} + \bar{P}(t) \right)$$

$$\mu \frac{d\bar{I}(t)}{dt} = -\bar{I}(t) + (1 - \bar{I}(t)) S_i \left(w_{ie} \bar{I}(t) - w_{ii} \bar{I}(t) + \bar{Q}(t) \right)$$

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

Spatially extended

$$\mu \frac{\partial \bar{E}(\mathbf{x}, t)}{\partial t} = -\bar{E}(\mathbf{x}, t) + (1 - \bar{E}(\mathbf{x}, t)) S_e \left((\bar{E} \otimes \omega_{ee})(\mathbf{x}, t) - (\bar{I} \otimes \omega_{ei})(\mathbf{x}, t) + \bar{P}(\mathbf{x}, t) \right)$$

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Spatially extended Wilson-Cowan model

Where do these equations come from?

Wilson-Cowan, Kybernetik, 1973

$$\mu \frac{\partial \bar{E}(x, t)}{\partial t} = -\bar{E}(x, t) + (1 - \bar{E}(x, t)) S_e \left((\bar{E} \circledast \omega_{ee})(x, t) - (\bar{I} \circledast \omega_{ei})(x, t) + \bar{P}(x, t) \right)$$

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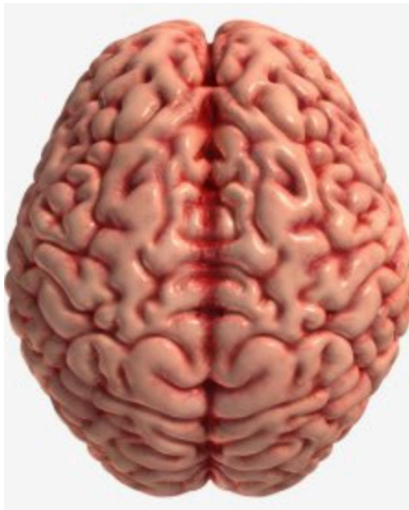
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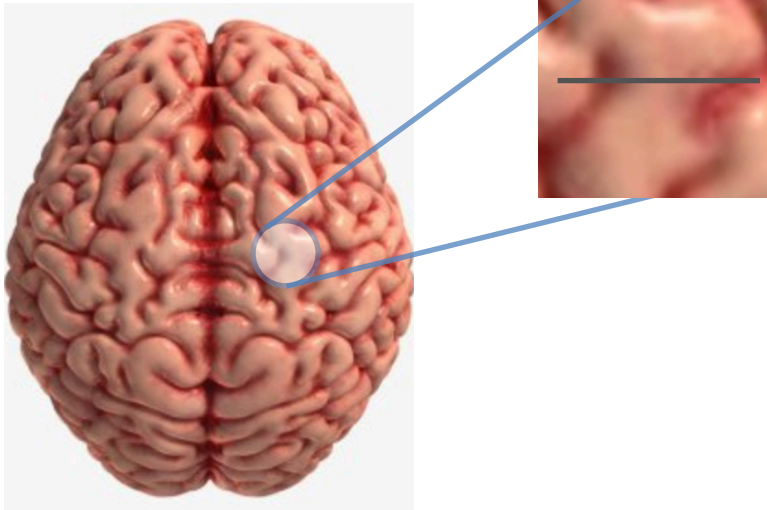
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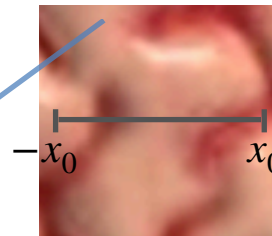
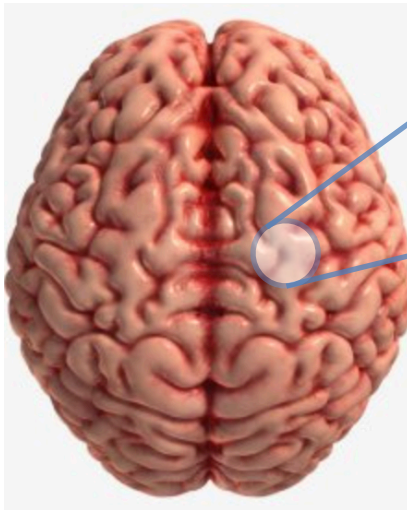
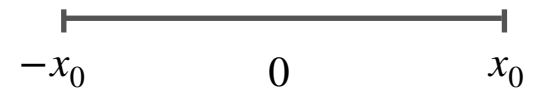
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- Cortical surface is assumed to be isotropic:
 - Direction has no impact on the connectivity,
 - Only distance matters in the connectivity.

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

$E(x, t)$: proportion of firing excitatory neurons at instant t and at position x

$I(x, t)$: proportion of firing inhibitory neurons at instant t and at position x

Attention! $E(x, t)$ and $\bar{E}(x, t)$ are not the same thing! $I(x, t)$ and $\bar{I}(x, t)$ are not the same thing!

- $E(x, t) = 0, I(x, t) = 0$ denote the resting state (low-level background neural activity).
 - Resting state appears when there is no synaptic or external stimulation.
- In some variants of the model, negative values of $E(x, t)$ or $I(x, t)$ are attained.
 - This represents depression of resting activity.
- $E(x, t)$ and $I(x, t)$ are referred to as excitatory and inhibitory neural activity, respectively.
 - Referred to as such, it should be tacitly understood that they correspond to the proportions of firing neurons.

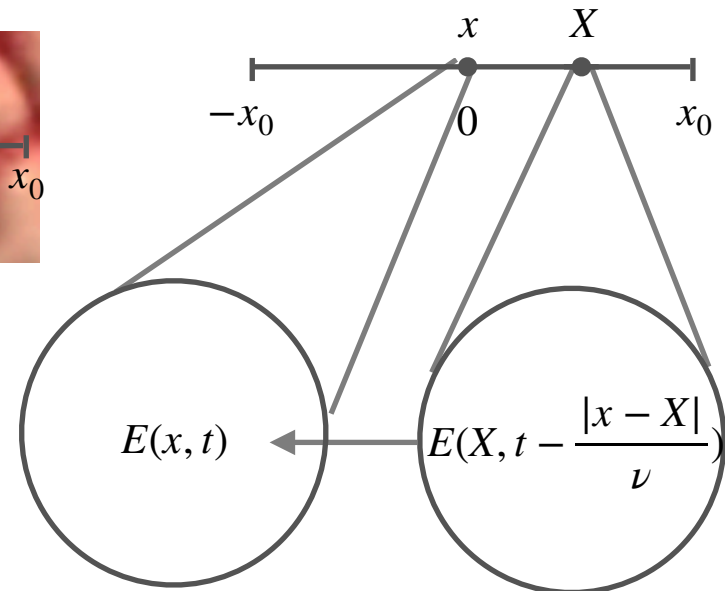
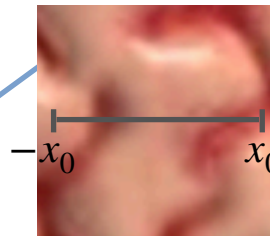
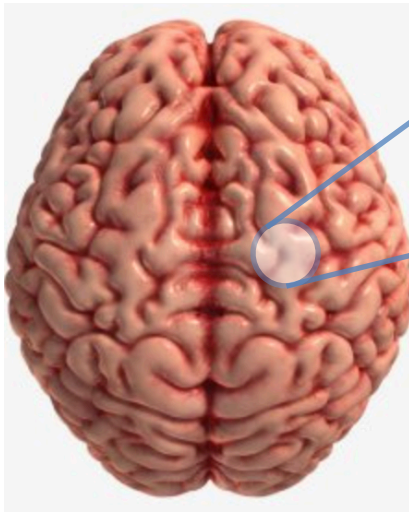
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Wilson-Cowan, Kybernetik, 1973

$E(x, t)$: proportion of firing excitatory neurons at instant t and at position x

$I(x, t)$: proportion of firing inhibitory neurons at instant t and at position x

- First, we consider only the excitatory neurons at x and at the instant t .
- We consider the arrival of average synaptic input to the excitatory neurons located at x , and caused by the activity of the excitatory neurons located at X and at the instant $t - \frac{|x - X|}{\nu}$. Here ν denotes the velocity of spike propagation.



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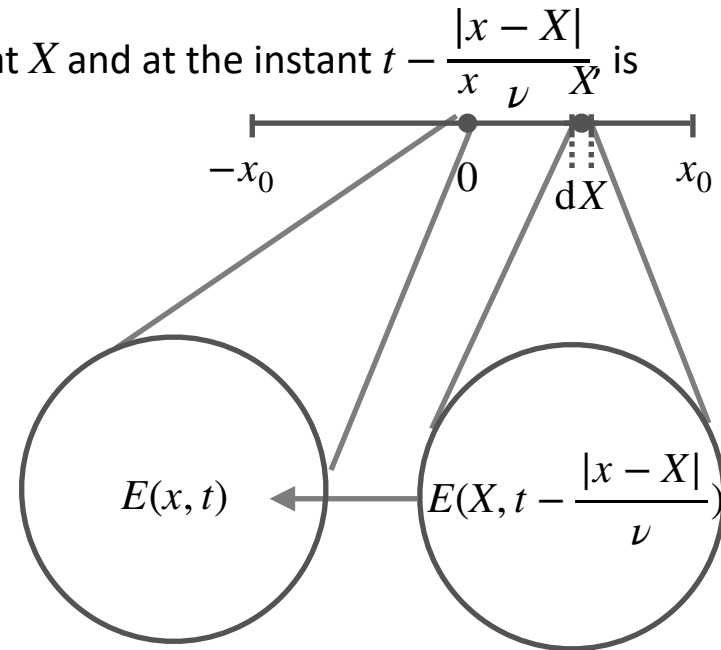
- We denote the velocity of spike propagation by ν .
- Then, the average synaptic input to the excitatory neurons located at x , and caused by the activity of the excitatory neurons located at X and at the instant $t - \frac{|x - X|}{\nu}$ is

written as:

$$E\left(X, t - \frac{|x - X|}{\nu}\right) \omega_{ee}(x - X) dX$$

$\downarrow$ proportion of the active exc. neurons at X and at the instant $t - \frac{|x - X|}{\nu}$.
 $\downarrow$ connectivity kernel: e $\rightarrow$ e
 $\uparrow$ infinitesimal length

delay due to the activity propagation



Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

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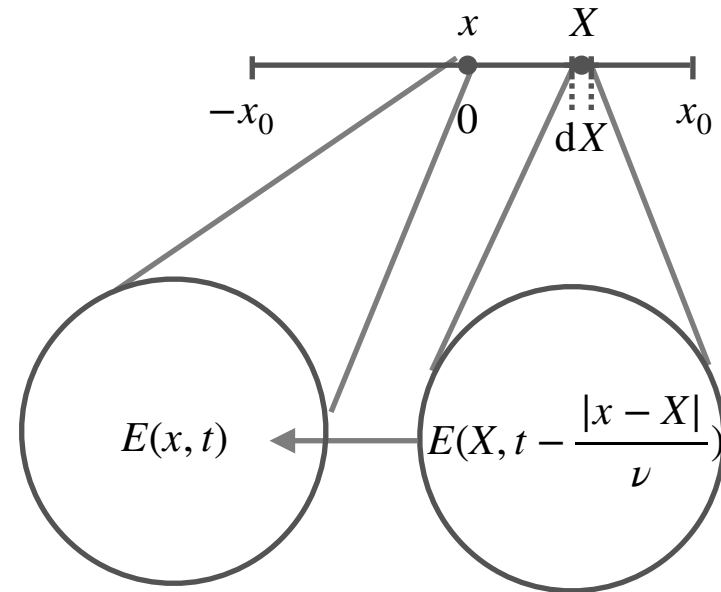
- We denote the velocity of spike propagation by ν .
- Then, the average synaptic input to the excitatory neurons located at x , and caused by the activity of the excitatory and inhibitory neurons located at X and at the instant

$t - \frac{|x - X|}{\nu}$, is written as:

$$E\left(X, t - \frac{|x - X|}{\nu}\right) \omega_{ee}(x - X) dX$$

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↑ infinitesimal length



$$I\left(X, t - \frac{|x - X|}{\nu}\right) \omega_{ei}(x - X) dX$$

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

$E(x, t)$: proportion of firing excitatory neurons at instant t and at position x

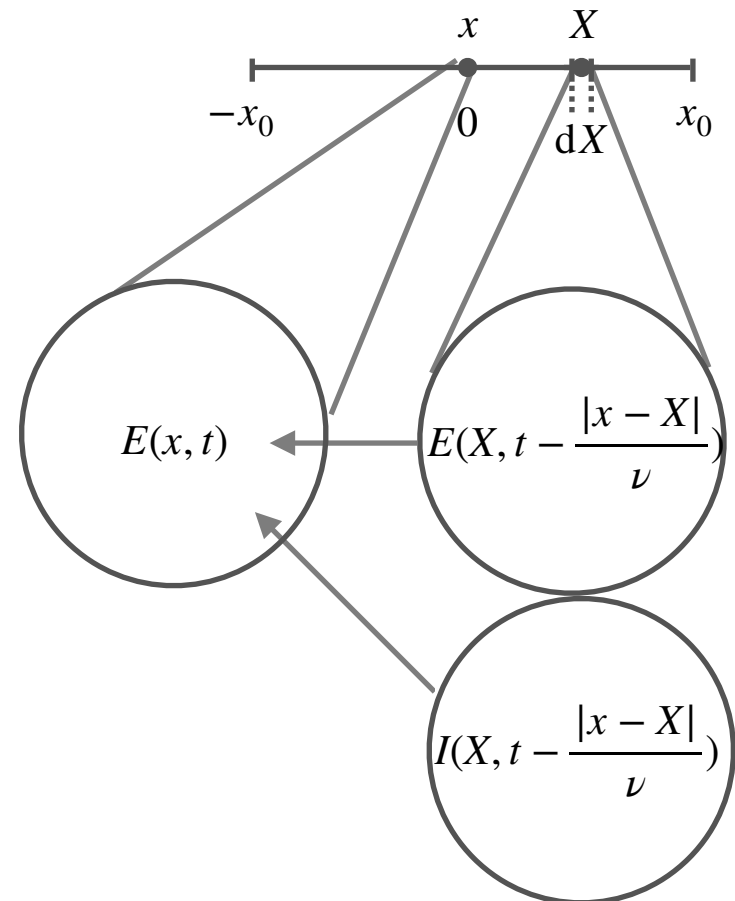
$I(x, t)$: proportion of firing inhibitory neurons at instant t and at position x

$$E\left(X, t - \frac{|x - X|}{\nu}\right) \omega_{ee}(x - X) dX$$

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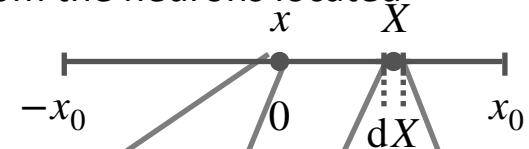
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- However, this is for the arrival of average synaptic input coming only from the neurons located at X !!! What about the synaptic input from the neurons located elsewhere on the cortical layer?

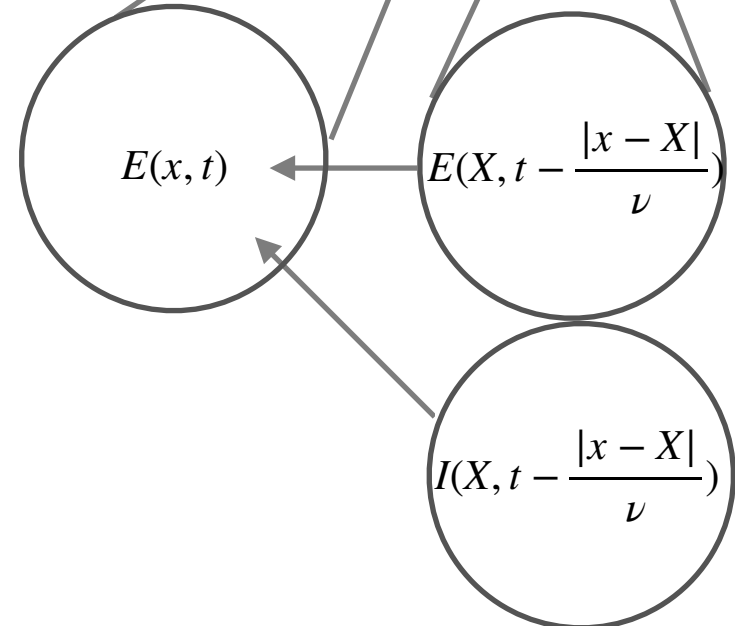


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Wilson-Cowan, Kybernetik, 1973

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$I(x, t)$: proportion of firing inhibitory neurons at instant t and at position x

- To find the average synaptic input from all excitatory and inhibitory neurons in the cortical tissue at time $t - |x - X|/\nu$, we use the following integrals (convolutions), respectively:

$$\int_{-x_0}^{x_0} E\left(X, t - \frac{|x - X|}{\nu}\right) \omega_{ee}(x - X) dX, \quad (\text{from excitatory neurons})$$

$$\int_{-x_0}^{x_0} I\left(X, t - \frac{|x - X|}{\nu}\right) \omega_{ei}(x - X) dX. \quad (\text{from inhibitory neurons})$$

- The first provides the synaptic input arriving at the instant t to the excitatory neurons located at x , caused by the activity of all the excitatory neurons on the cortical tissue at the instant $t - \frac{|x - X|}{\nu}$.
- The second is the same but now caused by the activity of all the inhibitory neurons on the cortical tissue at the instant $t - \frac{|x - X|}{\nu}$.
- It is plausible to ignore the delay $\frac{|x - X|}{\nu}$ since the propagation is very fast (i.e, ν is very large).

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$$\int_{-x_0}^{x_0} I(X, t) \omega_{ei}(x - X) dX. \quad (\text{from inhibitory neurons})$$

- Recall that individual neurons sum (integrate) their post-synaptic potentials (EPSPs + IPSPs) also in time and the result constitutes the synaptic input (see Slides_WilsonCowanLocalized). This input decays in time after it arrives to the target (receiving) neuron. We represent this decay with a function $\alpha(t)$. Then, the total level of synaptic input $\xi(x, t)$ to the excitatory neurons located at x and at time t will be:

$$\xi(x, t) = \int_{-\infty}^t \left[\int_{-x_0}^{x_0} E(X, t') \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} I(X, t') \omega_{ei}(x - X) dX + P(x, t') \right] \alpha(t - t') dt'.$$

- A plausible choice for the decay function is $\alpha(t - t') := e^{\frac{t-t'}{\mu}}$.

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

$E(x, t)$: proportion of firing excitatory neurons at instant t and at position x

$I(x, t)$: proportion of firing inhibitory neurons at instant t and at position x

- Transfer functions $S_e(\xi)$ and $S_i(\xi)$ give the proportion of neurons which gets activated (fires) with respect to the synaptic input ξ that excitatory and inhibitory populations receives, respectively, **if none of the neurons were initially in the refractory period**.
- The proportion of excitatory neurons which are non-refractory at x and at time t is:

$$1 - \int_{t-r}^t E(x, t') dt',$$

where r denotes the duration of refractory period. It is similar for the inhibitory neurons.

- Finally, the proportion of activated excitatory neurons at x , and at time $t + \mu$, where the activation delay is equal to μ , is described by

$$E(x, t + \mu) = \left(1 - \int_{t-r}^t E(x, t') dt'\right) S_e \left[\int_{-\infty}^t \left(\int_{-x_0}^{x_0} E(X, t') \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} I(X, t') \omega_{ei}(x - X) dX + P(x, t') \right) \alpha(t - t') dt' \right].$$

- The same procedure provides the following expression for the activated inhibitory neurons as well:

$$I(x, t + \mu) = \left(1 - \int_{t-r}^t I(x, t') dt'\right) S_i \left[\int_{-\infty}^t \left(\int_{-x_0}^{x_0} E(X, t') \omega_{ie}(x - X) dX - \int_{-x_0}^{x_0} I(X, t') \omega_{ii}(x - X) dX + Q(x, t') \right) \alpha(t - t') dt' \right].$$

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

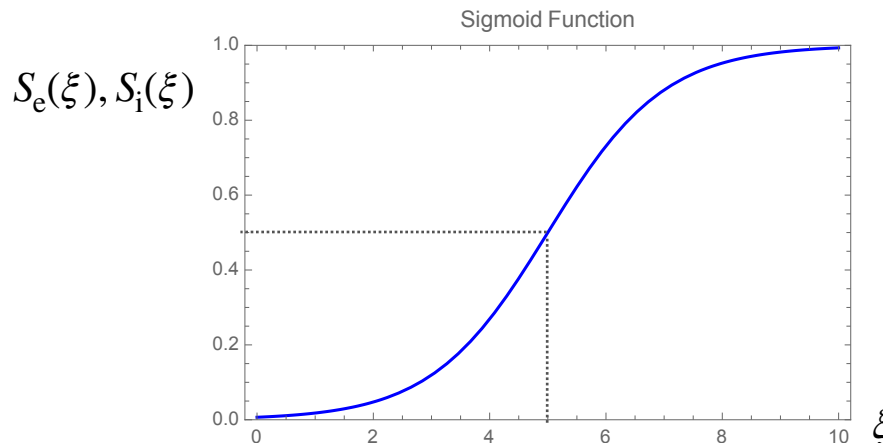
$E(x, t)$: proportion of firing excitatory neurons at instant t and at position x

$I(x, t)$: proportion of firing inhibitory neurons at instant t and at position x

$$E(x, t + \mu) = \left(1 - \int_{t-r}^t E(x, t') dt'\right) S_e \left[\int_{-\infty}^t \left(\int_{-x_0}^{x_0} E(X, t') \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} I(X, t') \omega_{ei}(x - X) dX + P(x, t') \right) \alpha(t - t') dt' \right]$$

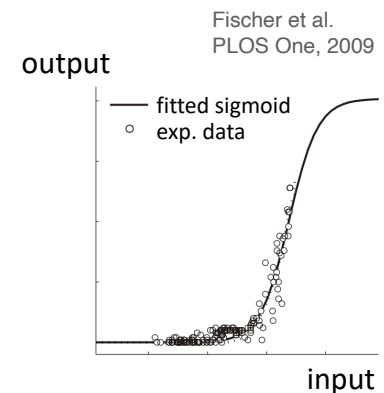
$$I(x, t + \mu) = \left(1 - \int_{t-r}^t I(x, t') dt'\right) S_i \left[\int_{-\infty}^t \left(\int_{-x_0}^{x_0} E(X, t') \omega_{ie}(x - X) dX - \int_{-x_0}^{x_0} I(X, t') \omega_{ii}(x - X) dX + Q(x, t') \right) \alpha(t - t') dt' \right]$$

- A typical choice for the transfer functions is the sigmoid, as in localized Wilson-Cowan model:



Why sigmoid?

- Low excitation → no neurons activated: sigmoid models well.
- High excitation → all neurons activated: sigmoid models well.
- Experimental evidence: sigmoid-like input-output curves observed experimentally in neurons.



Spatially extended Wilson-Cowan model

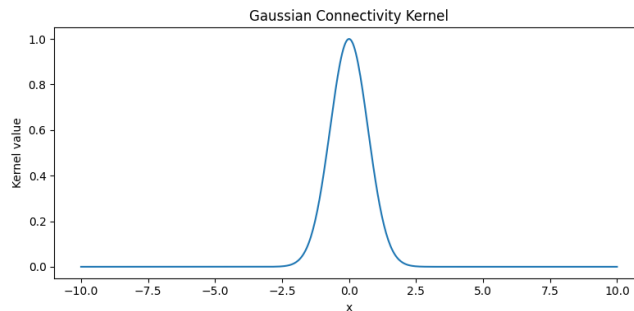
Wilson-Cowan, Kybernetik, 1973

$E(x, t)$: proportion of firing excitatory neurons at instant t and at position x

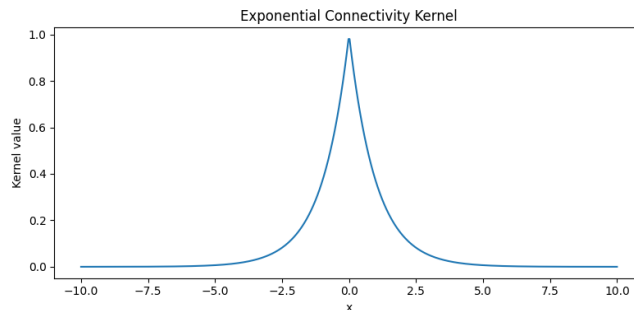
$I(x, t)$: proportion of firing inhibitory neurons at instant t and at position x

$$E(x, t + \mu) = \left(1 - \int_{t-r}^t E(x, t') dt'\right) S_e \left[\int_{-\infty}^t \left(\int_{-x_0}^{x_0} E(X, t') \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} I(X, t') \omega_{ei}(x - X) dX + P(x, t') \right) \alpha(t - t') dt' \right]$$

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$$\omega(x) := e^{-\frac{|x|^2}{\sigma}}$$



$$\omega(x) := e^{-\frac{|x|}{\sigma}}$$

Spatially extended Wilson-Cowan model

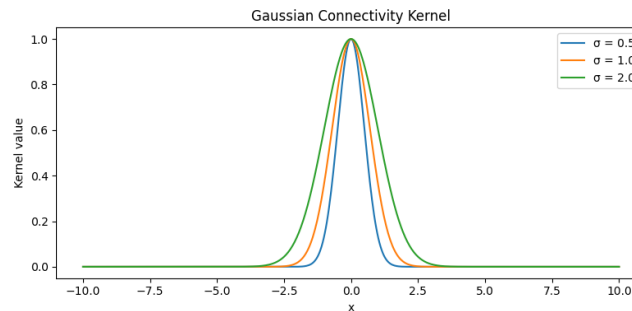
Wilson-Cowan, Kybernetik, 1973

$E(x, t)$: proportion of firing excitatory neurons at instant t and at position x

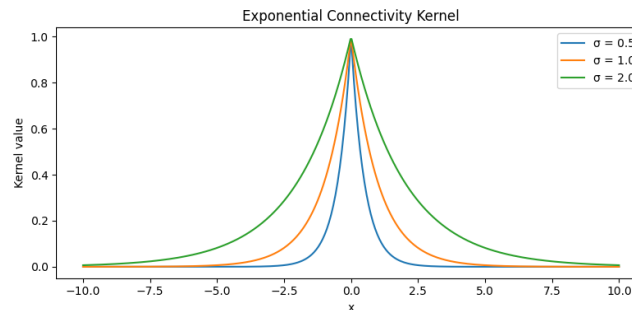
$I(x, t)$: proportion of firing inhibitory neurons at instant t and at position x

$$E(x, t + \mu) = \left(1 - \int_{t-r}^t E(x, t') dt'\right) S_e \left[\int_{-\infty}^t \left(\int_{-x_0}^{x_0} E(X, t') \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} I(X, t') \omega_{ei}(x - X) dX + P(x, t') \right) \alpha(t - t') dt' \right]$$

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$$\omega(x) := e^{-\frac{|x|}{\sigma}}$$

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

(E1)

$$E(x, t + \mu) = \left(1 - \int_{t-r}^t E(x, t') dt'\right) S_e \left[\int_{-\infty}^t \left(\int_{-x_0}^{x_0} E(X, t') \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} I(X, t') \omega_{ei}(x - X) dX + P(x, t') \right) \alpha(t - t') dt' \right]$$

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Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

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- This system provides a detailed representation of neural population activity. However, it is complicated to study and analyze. So, we simplify this system by using the time averages $\bar{E}(x, t)$ and $\bar{I}(x, t)$ of $E(x, t)$ and $I(x, t)$, respectively:

$$\bar{E}(x, t) := \frac{1}{\mu} \int_{-\infty}^t E(x, t') \alpha(t - t') dt', \quad \bar{I}(x, t) := \frac{1}{\mu} \int_{-\infty}^t I(x, t') \alpha(t - t') dt'.$$

It is the same for $\bar{P}(x, t)$ and $\bar{Q}(x, t)$. Here, recall that μ denotes the time constant.

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

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- A typical (and adequate) choice for the decay $\alpha(t)$ of the neural inputs is the exponential decay function, i.e., $\alpha(t) = e^{-t/\mu}$. The time averages become:

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Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

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- We can write (E1) by using the time averages as follows:

$$E(x, t + \mu) = \left(1 - \int_{t-r}^t E(x, t') dt'\right) S_e \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t) \omega_{ei}(x - X) dX + \mu \bar{P}(x, t) \right],$$

$$I(x, t + \mu) = \left(1 - \int_{t-r}^t I(x, t') dt'\right) S_i \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ie}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t) \omega_{ii}(x - X) dX + \mu \bar{Q}(x, t) \right].$$

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

(E2)

$$E(x, t + \mu) = \left(1 - \int_{t-r}^t E(x, t') dt'\right) S_e \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t) \omega_{ei}(x - X) dX + \mu \bar{P}(x, t) \right]$$

$$I(x, t + \mu) = \left(1 - \int_{t-r}^t I(x, t') dt'\right) S_i \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ie}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t) \omega_{ii}(x - X) dX + \mu \bar{Q}(x, t) \right]$$

- A second simplification is made on the integrals regarding the proportion of nonrefractory neurons in the populations by using the time derivatives of $\bar{E}(x, t)$ and $\bar{I}(x, t)$:

$$E(x, t) = \mu \frac{\partial}{\partial t} \bar{E}(x, t) + \bar{E}(x, t), \quad I(x, t) = \mu \frac{\partial}{\partial t} \bar{I}(x, t) + \bar{I}(x, t). \quad (\text{Leibniz integral rule})$$

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

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- The refractory period r is much smaller than the time constant μ . Therefore, we may use the following approximations:

$$\bar{E}(x, t - r) \approx \bar{E}(x, t), \quad \bar{I}(x, t - r) \approx \bar{I}(x, t), \quad \implies \int_{t-r}^t E(x, t') dt' \approx r \bar{E}(x, t), \quad \int_{t-r}^t I(x, t') dt' \approx r \bar{I}(x, t),$$

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

(E3)

$$E(x, t + \mu) = \left(1 - r \bar{E}(x, t)\right) S_e \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t') \omega_{ei}(x - X) dX + \mu \bar{P}(x, t) \right]$$

$$I(x, t + \mu) = \left(1 - r \bar{I}(x, t)\right) S_i \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ie}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t') \omega_{ii}(x - X) dX + \mu \bar{Q}(x, t) \right]$$

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Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

(E3)

$$E(x, t + \mu) = \left(1 - r \bar{E}(x, t)\right) S_e \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t') \omega_{ei}(x - X) dX + \mu \bar{P}(x, t) \right]$$

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which provide (E3).

- Similarly to localized Wilson-Cowan, we can use the Taylor expansion and ignore the high frequency components of E and I :

$$E(x, t + \mu) \approx \mu \frac{\partial}{\partial t} \bar{E}(x, t) + \bar{E}(x, t), \quad I(x, t + \mu) \approx \mu \frac{\partial}{\partial t} \bar{I}(x, t) + \bar{I}(x, t).$$

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

(E3)

$$E(x, t + \mu) = \left(1 - r \bar{E}(x, t)\right) S_e \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t) \omega_{ei}(x - X) dX + \mu \bar{P}(x, t) \right]$$

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Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

(E4)

$$\mu \frac{\partial}{\partial t} \bar{E}(x, t) = -\bar{E}(x, t) + \left(1 - r \bar{E}(x, t)\right) S_e \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ee}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t) \omega_{ei}(x - X) dX + \mu \bar{P}(x, t) \right]$$

$$\mu \frac{\partial}{\partial t} \bar{I}(x, t) = -\bar{I}(x, t) + \left(1 - r \bar{I}(x, t)\right) S_i \left[\int_{-x_0}^{x_0} \mu \bar{E}(X, t) \omega_{ie}(x - X) dX - \int_{-x_0}^{x_0} \mu \bar{I}(X, t) \omega_{ii}(x - X) dX + \mu \bar{Q}(x, t) \right]$$

- A second simplification is made on the integrals regarding the proportion of non refractory neurons in the populations by using the time derivatives of $\bar{E}(x, t)$ and $\bar{I}(x, t)$:

$$E(x, t) = \mu \frac{\partial}{\partial t} \bar{E}(x, t) + \bar{E}(x, t), \quad I(x, t) = \mu \frac{\partial}{\partial t} \bar{I}(x, t) + \bar{I}(x, t). \quad (\text{Leibniz integral rule})$$

- The refractory period r is much smaller than the time constant μ . Therefore, we may use the following approximations:

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which provide (E3).

- Similarly to localized Wilson-Cowan, we can use the Taylor expansion and ignore the high frequency components of E and I :

$$E(x, t + \mu) \approx \mu \frac{\partial}{\partial t} \bar{E}(x, t) + \bar{E}(x, t), \quad I(x, t + \mu) \approx \mu \frac{\partial}{\partial t} \bar{I}(x, t) + \bar{I}(x, t).$$

- This gives us (E4).

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

(E5)

$$\mu \frac{\partial}{\partial t} \bar{E}(x, t) = -\bar{E}(x, t) + (1 - \bar{E}(x, t)) S_e \left((\bar{E} \otimes \omega_{ee})(x, t) - (\bar{I} \otimes \omega_{ei})(x, t) + \bar{P}(x, t) \right)$$

$$\mu \frac{\partial}{\partial t} \bar{I}(x, t) = -\bar{I}(x, t) + (1 - \bar{I}(x, t)) S_i \left((\bar{E} \otimes \omega_{ie})(x, t) - (\bar{I} \otimes \omega_{ii})(x, t) + \bar{Q}(x, t) \right)$$

- A second simplification is made on the integrals regarding the proportion of non refractory neurons in the populations by using the time derivatives of $\bar{E}(x, t)$ and $\bar{I}(x, t)$:

$$E(x, t) = \mu \frac{\partial}{\partial t} \bar{E}(x, t) + \bar{E}(x, t), \quad I(x, t) = \mu \frac{\partial}{\partial t} \bar{I}(x, t) + \bar{I}(x, t). \quad (\text{Leibniz integral rule})$$

- The refractory period r is much smaller than the time constant μ . Therefore, we may use the following approximations:

$$\bar{E}(x, t - r) \approx \bar{E}(x, t), \quad \bar{I}(x, t - r) \approx \bar{I}(x, t), \quad \implies \int_{t-r}^t E(x, t') dt' \approx r \bar{E}(x, t), \quad \int_{t-r}^t I(x, t') dt' \approx r \bar{I}(x, t),$$

which provide (E3).

- Similarly to localized Wilson-Cowan, we can use the Taylor expansion and ignore the high frequency components of E and I :

$$E(x, t + \mu) \approx \mu \frac{\partial}{\partial t} \bar{E}(x, t) + \bar{E}(x, t), \quad I(x, t + \mu) \approx \mu \frac{\partial}{\partial t} \bar{I}(x, t) + \bar{I}(x, t).$$

- This gives us (E4).
- Finally, we can include μ in the connectivity kernels and write the convolutions by using the notation $\otimes$. This gives the final form of the spatially extended Wilson-Cowan model, (E5). 41

Spatially extended Wilson-Cowan model

Wilson-Cowan, Kybernetik, 1973

$$\mu \frac{\partial}{\partial t} \bar{E}(x, t) = -\bar{E}(x, t) + (1 - \bar{E}(x, t)) S_e \left((\bar{E} \otimes \omega_{ee})(x, t) - (\bar{I} \otimes \omega_{ei})(x, t) + \bar{P}(x, t) \right)$$

$$\mu \frac{\partial}{\partial t} \bar{I}(x, t) = -\bar{I}(x, t) + (1 - \bar{I}(x, t)) S_i \left((\bar{E} \otimes \omega_{ie})(x, t) - (\bar{I} \otimes \omega_{ii})(x, t) + \bar{Q}(x, t) \right)$$

- Active transients
- Spatially localized limit cycles (spatially localized tonic spiking)
- Travelling waves

See the animations!

The end